

Assignment 1: Computers in the Past, Present and Future

Deadline: Next Friday

Introduction

Computers have changed enormously since they were first built. They went from giant machines used by a few scientists to devices almost everyone carries in their pocket. This assignment looks at how computers were in the past, how they are today, and how they may be in the future.

Past (1940s to 1980s)

The first computers, like ENIAC, were as big as a room. They were built with vacuum tubes and later transistors. Input was done with punch cards, and only trained experts could run them. By the 1970s and 80s, microprocessors led to the first personal computers, such as the IBM PC.

Advantages

- Did calculations far faster than humans
- Became the foundation for all modern computing
- Useful for military work, census data and science

Disadvantages

- Enormous, very expensive and slow by today's standards
- Used a lot of electricity and produced heat
- Broke down often and had very little memory

Present (1990s to Now)

Today we have laptops, smartphones, tablets, smartwatches and cloud computing. The internet connects them all, and AI is built into search, maps, phones and apps.

Advantages

- Fast, cheap, portable and easy to use
- Instant communication, online learning, banking, shopping and entertainment
- Allows remote work and automates boring tasks

Disadvantages

- Privacy and data theft are real risks, along with hacking and viruses
- Screen addiction and less face-to-face time can hurt health and relationships
- E-waste is a growing problem, and some jobs are being replaced by automation

Future

Likely developments include quantum computers, more powerful AI, wearable devices like smart glasses, and brain-computer interfaces. Computers may also run on far less energy.

Advantages

- Faster medical research and disease prediction
- Better climate modelling and more personalised education
- Smart cities and self-driving transport

Disadvantages

- Many jobs may be lost to automation
- Powerful AI and quantum machines could be misused, for example to break today's encryption
- Constant data collection raises privacy fears, and people without access may fall behind

Comparison: Computers in Past, Present and Future

Feature	Past (1940s-1980s)	Present (1990s-Now)	Future
Size	Room-sized	Pocket to desktop	Wearable, tiny
Speed	Thousands of operations per second	Billions per second	Potentially far beyond this (quantum)
	Only governments	Affordable for	Probably cheaper, but

Cost	and universities	most people	advanced tech may start expensive
Users	Expert scientists	Almost everyone	Everyone, often hands-free by voice
Technology	Vacuum tubes, transistors	Microchips, internet, cloud, AI	Quantum, advanced AI, biotech links
Advantages	Fast calculations, foundation for modern computing	Portable, cheap, easy to use, connects the world	Faster research, smart cities, personalised education
Disadvantages	Huge, costly, slow, high power use, frequent failures	Privacy risks, hacking, screen addiction, e-waste	Job loss, misuse of AI, privacy concerns, digital divide
Main risk	Failures and high cost	Privacy and addiction	Job loss and misuse of AI

Timeline

Past (1940s-1980s) --> Present (1990s-Now) --> Future

Huge, slow machines Personal, mobile, cloud and AI Quantum, smarter AI, wearable

Conclusion

Each stage solved the problems of the one before it. Computers became smaller, faster and cheaper, but new issues like privacy and job change appeared along the way. The future depends on using them responsibly.